

PANDA Component Description: *timer_16bit*

Overview

The *timer_16bit* component implements a 16-bit timer with adjustable prescaler and user settable auto reload value. The timer can be configured through a configuration register which controls the following options: run/stop, up/down-counter, auto reload enable and interrupt enable. The timer is set and read over a shadow register to provide consistent data over both 8-bit parts of this 16-bit value.

The prescaler is directly calculated from the two generics. Therefore the user has only to know the clock frequency and the desired timer resolution.

Generics

Name	Type	Default	Description
clk_frequency	integer	10'000'000	Clock frequency in Hz
resolution_us	integer	1'000	Timer Resolution in μ s (1...1'000'000)

Registers

Timer Value Registers

Name: TMRL

Address: Base + 0x00

Name	TMRL
R/W	R/W
Reset Value	0x00

Name: TMRH

Address: Base + 0x01

Name	TMRH
R/W	R/W
Reset Value	0x00

TMRH and *TMRL* represent together the 16-bit timer value. *TMRH* holds the high byte, *TMRL* the low byte. On write operations on the *SHAD* register, the *TMRH/TMRL* value is written to the timer. On read operations on *SHAD* *TMRH/TMRL* are updated with the current timer value. This provides consistent 16-bit data even if only 8-bit registers can be read at once. Without operation on *SHAD* the *TMRH/TMRL* value remains unchanged even if the timer runs.

Reload Registers

Name: RLDL

Address: Base + 0x02

Name	RLDL
R/W	R/W
Reset Value	0x00

Name: RLDH

Address: Base + 0x03

Name	RLDH
R/W	R/W
Reset Value	0x00

RLDH and *RLDL* represent together a 16-bit reload value. This value is loaded into the timer if an overflow occurs and the autoreload option is enabled (*RELOAD* in *CFG* is set).

Shadow Register

Name: SHAD

Address: Base + 0x04

Name	SHAD
R/W	-
Reset Value	-

The *SHAD* register is not a real register, it only represents the command to read/write the timer from/to *TMRL/TMRH*. If *SHAD* is read, *TMRL/TMRH* are loaded with the current timer value. If *SHAD* is written, the timer is loaded with the *TMRL/TMRH* value. The *SHAD* register returns always 0x00 on read operations.

Configuration Register

Name: CFG

Address: Base + 0x05

Name	-	-	-	-	UPDOWN	RELOAD	RUN	INTON
R/W	-	-	-	-	R/W	R/W	R/W	R/W
Reset Value	0	0	0	0	0	0	0	0

INTON: '1' → State of *INT* is forwarded to the interrupt line
 '0' → No interrupts are generated

RUN: '1' → Timer runs
 '0' → Timer is stopped

RELOAD: '1' → The timer is loaded with the value of *RLD* after an overflow
 '0' → The timer is not changed after an overflow

UPDOWN: '1' → Timer runs up, overflow from 0xFFFF to 0x0000
 '0' → Timer runs down, overflow from 0x0000 to 0xFFFF

Interrupt Register

Name: INT

Address: Base + 0x06

Name	INT						
R/W	-	-	-	-	-	-	R/W
Reset Value	-	-	-	-	-	-	0

INT is set on every overflow of the timer regardless of the state of *INTON* in *CFG*. It must be cleared by Software.

Unused Register

Address: Base + 0x07

This register does not exist. It is always read as 0x00 and write operations do not have any effect.

Driver Macros

All 16-bit variables must hold their LSB in the lower address and the MSB in the higher address to provide proper function of the driver macros.

START

Call: *_START

Starts the timer (sets *RUN* in *CFG*).

STOP

Call: *_STOP

Stops the timer (clears *RUN* in *CFG*).

CLRINT

Call: *_CLRINT

Clears the interrupt (clears *INT*).

READTMR

Call: *_READTMR [variable]

Parameter	Description	Immediate/Variable	Size
variable	Destination for <i>TMR</i> value	Variable	16-Bit

Reads the timer over the shadow register and stores the its value into the variable.

SETTMR

Call: *_SETTMR [value]

Parameter	Description	Immediate/Variable	Size
value	New <i>TMR</i> value	Immediate/Variable	16-Bit

Sets the timer over the shadow register.

INIT_CONTINT

Call: *_INIT_CONTINT [intervall]

Parameter	Description	Immediate/Variable	Size
intervall	Interrupt intervall in ticks	Immediate	16-Bit

Initializes the timer for continuous interrupt generation. The general IE bit of the CPU must be set by user software.

INIT_WAITNTICKS

Call: *_INIT_WAITNTICKS

Initializes the timer for the usage of the WAITNTICKS macro.

WAITNTICKS

Call: *_WAITNTICKS [N]

Parameter	Description	Immediate/Variable	Size
N	Delay time in ticks	Immediate/Variable	16-Bit

Waits for the specified time. This macro can replace delay loops.